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Network Traffic Anomaly Detection and Malicious Behavior Simulation Analysis Based on Deep Q Network and Self-encoder

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ABSTRACT

The currently designed network traffic anomaly detection and malicious behavior matrix are generally unidirectional, with a small traffic recognition range, resulting in a decrease in the final F1 value. Therefore, this article proposes a network traffic anomaly detection and malicious behavior simulation analysis method that combines deep Q-network and autoencoder. Reduced the dimensionality of network traffic data and expanded the range of traffic recognition in multiple steps; Design a multi-step network traffic anomaly detection matrix and construct a network traffic anomaly detection model that combines deep Q-network and autoencoder. Using malicious behavior recognition and abnormal traffic locking correction to achieve simulation processing. The experimental results indicate that by comparing the seven selected traffic operation cycles, the final F1 value can reach 0.8, which is relatively close to 1. This indicates that combining deep Q networks and self coding technology, even in malicious attack control environments, the detection of abnormal traffic problems is relatively accurate, with controllable errors, and has practical application value.

CCS CONCEPTS

• **Security and privacy** → Network security; Denial-of-service attacks.

KEYWORDS

Fusion depth Q network, Self-encoder, Network traffic, Anomaly detection

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1 INTRODUCTION

The comprehensive coverage of the Internet has brought convenience to production and daily life [1], but it also brings the risk of abnormal network traffic. Therefore, it is necessary to strengthen the monitoring and control of network traffic to prevent viruses, network attacks, and spam from harming the network environment and traffic [2]. Network intrusion attacks are one of the important reasons for abnormal traffic, posing a serious threat to the security of users' personal information. Search engine anomalies, regional data anomalies, and server anomalies are common causes of traffic anomalies. These anomalies have various forms and are difficult to capture and identify during normal operation [3-5] .

In order to solve the above problems, relevant personnel have designed the corresponding network traffic anomaly monitoring and malicious behavior identification methods. Y Jiang et al. in order to enhance data transmission security in the context of Industry 4.0, propose an integrated data-driven detection framework for the control layer, which can handle secure transmission and attack detection simultaneously. A correlation-based secure encryption and decryption method is proposed to analyze the attack sensitivity in depth, which effectively improves the security of power grid data transmission [6]. Canavese D et al. propose a plaintext encrypted communication method based on machine learning techniques to monitor the kinds of tools that generate analyzed traffic by analyzing the comprehensive performance of different classifiers, which is helpful to improve the detection efficiency of early distributed denial of service attacks [7]. However, the above approach is unidirectional to the overall structure of the framework. Although it can achieve the expected tasks and objectives, it is difficult to control, and it is often affected by external environment and specific factors, resulting in errors or defects in the final detection results. Moreover, the single network traffic anomaly monitoring and malicious behavior identification method is inefficient and lacks pertinence and stability, which is one of the main reasons for the inaccurate and reliable detection results [8-10] . For this reason, this paper proposes a network traffic anomaly detection and malicious behavior simulation analysis based on deep Q network and self-encoder.

2 DESIGN THE METHOD OF ANOMALY DETECTION AND MALICIOUS BEHAVIOR SIMULATION BASED ON NETWORK TRAFFIC FUSION DEPTH Q NETWORK AND SELF-ENCODER

2.1 Dimensionality reduction of network traffic data

The basic characteristics of network traffic data are extracted by data preprocessing to ensure the reliability of abnormal traffic detection results. At this time, the characteristics of network traffic data are high-dimensional, so it is necessary to integrate deep Q-networks,

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and reduce the dimension of traffic data in the form of (T-distributed random neighbor embedding T-SNE).

Convert the currently collected data information into low-dimensional data, divide the specific distribution positions of abnormal points, and determine the corresponding abnormal time. SplitCap files are used to segment traffic data while determining the characteristics of the formatted traffic data itself. However, the transformation form of network data is not fixed, and combined with the actual traffic state and traffic packet size, traffic data with the same characteristics are segmented together to form a stable dimensionality reduction processing set . At this point, measure and calculate the reduction of bytes in the flow dimension, as shown in formula 1:

$$H = o^2 + (1 - \pi) \times \upsilon \tag{1}$$

In Formula 1:Hrepresents traffic dimensionality reduction bytes,$o$represents a dimension reduction controllable value,πrepresents the conversion flow difference,υrepresents the mean value of directional integration. Combined with the current measurement, the calculation of flow dimension reduction bytes is completed. It is set as the standard value for dimensionality reduction of data traffic, and the conversion of traffic data is assisted by fusing deep Q network, which strengthens the control of data conversion error and lays the foundation for subsequent traffic anomaly detection.

2.2 Design multi-order network abnormal traffic detection matrix

Due to the huge amount of network information and data, traditional processing and detection methods can no longer meet practical needs. Therefore, an anomaly detection model EB-GAN framework is proposed, which is connected to the initial traffic detection program. Under the guidance of BiGAN, a multidimensional and specific combination anomaly detection mechanism is established. The improved mixed loss function at this time is determined to be 16.35-21.45 parentheses, and the actual multi-level network traffic anomaly detection matrix structure is established, as shown in Figure 1

According to Figure 1, the design and practical analysis of the multi-level network abnormal traffic detection matrix structure is completed. The currently designed detection matrix can process and calibrate the actual abnormal traffic and form a stable controllable traffic measurement data set. At this time, the encoder E, the generator G and the discriminator are connected to the current matrix at the same time, and multiple levels of traffic detection structures are set, and the relationship with the set targets is established, and the final detection data and information are obtained through matrix calculation. However, it should be noted that the multi-level abnormal traffic detection standard currently designed is not fixed, but changes accordingly with the actual data traffic situation, which makes the matrix more flexible and stable to some extent and strengthens the final abnormal traffic detection effect.

2.3 Build a fusion depth q network+self-encoder network traffic anomaly detection model

Combining the fusion of deep Q-network and self coding technology, a network traffic anomaly detection model target was constructed under the guidance of generating adversarial networks (GANs). The anomaly detection model is divided into three modules: traffic data preprocessing, traffic data resampling, and multi classifier anomaly detection. Based on the rules for setting anomaly detection conditions, a malicious attack recognition program needs to be set up in the model and integrated with the control structure. Through self coding assisted measurement, malicious attacks were identified for the first time, strengthening effective control over network operation and traffic anomalies. The design and verification application of a network traffic anomaly detection model based on deep Q networks and self encoders were completed.

2.4 Malicious behavior identification and abnormal traffic lock correction to achieve simulation processing

Combining deep Q network technology with self coding technology, a malicious behavior recognition program was designed to lock in abnormal traffic and complete simulation testing. Input the collected non digital feature traffic into the model, convert characters into digital features, construct malicious attack recognition features, and design a processing structure based on the processing requirements of abnormal traffic, as shown in Figure 2

According to Figure 2, the design of malicious behavior identification and abnormal traffic locking correction structure is completed. According to this structure, the target is executed, the control of network environment is strengthened, and the final detection result is strengthened.

3 EXPERIMENT

This time, it is mainly to analyze and verify the actual effect of network traffic anomaly detection and malicious behavior simulation which integrates deep Q network and self-encoder. Considering the authenticity and reliability of the final test results, the G network is selected for auxiliary test and verification analysis. Use professional equipment and devices to collect basic test data and information, and then collect and integrate them for subsequent use. Next, according to the current measurement requirements standards, combined with the fusion depth Q network and self-encoder technology, the initial test environment is built.

3.1 Experimental preparation

Combined with the fusion of deep Q-network and self-encoder technology, the selected Q-network traffic anomaly detection and malicious behavior simulation test environment is constructed and set. According to the specific situation of the current network operation, seven cycles are set, and each cycle needs to set a data acquisition time, which is convenient for the subsequent detection and directional analysis of abnormal traffic. Then, the original data set of network traffic is established every cycle, and the combination is completed after the integration of test set and training set, and the abnormal data is calibrated in the process. Set the sliding window size to 224, and the step size that the unit can recognize is 5.5. The SA-WGAN model is connected to the control program, and the set network traffic length is 116. The LSTM structure is used to associate and bind the generators A1 and A2, which is convenient for multidimensional detection of abnormal traffic. At this time, based on the current measurement requirements and standards, the

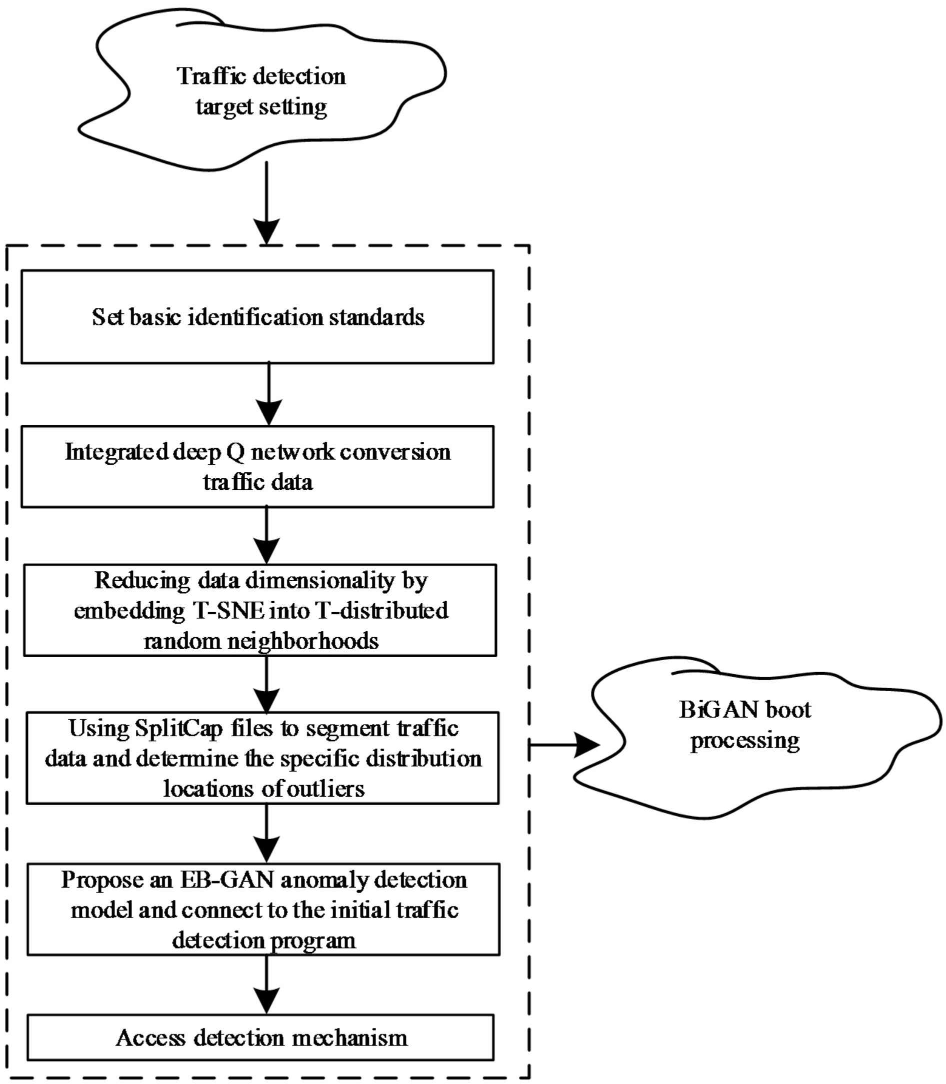

Figure 1: Structural diagram of abnormal traffic detection matrix of multi-level network.

hidden detection unit value is calculated, as shown in the following formula 2:

$$D = (1 + n)^2 \times \mu - \frac{1}{dr} \tag{2}$$

In Formula 2:Drepresents a hidden detection unit value,$n$represents the compensation difference,μrepresents the discriminant mean,$d$indicate that anomaly detection is time-consuming,$r$indicates the detection frequency. Combined with the current measurement, the calculation of hidden detection unit value is completed. Then, on this basis, a multi-level unit flow monitoring program is designed to form a directional integrated structure with the main control program, and the basic flow is identified from multiple angles and cycles, and then the basic test indicators and control parameters are set, as shown in the following table 1

According to Table 1, the basic test indicators and parameters are set, and then, based on this, combined with the currently set basic environment, an abnormal traffic detection environment and malicious behavior identification program with cyclic habits and stability are constructed. At this point, after the construction of the test background, next, combined with the fusion depth Q network and self-encoder technology, the corresponding measurement and verification analysis are carried out.

3.2 Experimental process and result analysis

In the above-mentioned test environment, combined with the fusion of deep Q network and self-encoder technology, the selected Q network traffic anomaly detection and malicious behavior simulation are tested and verified. At present, the basic test environment

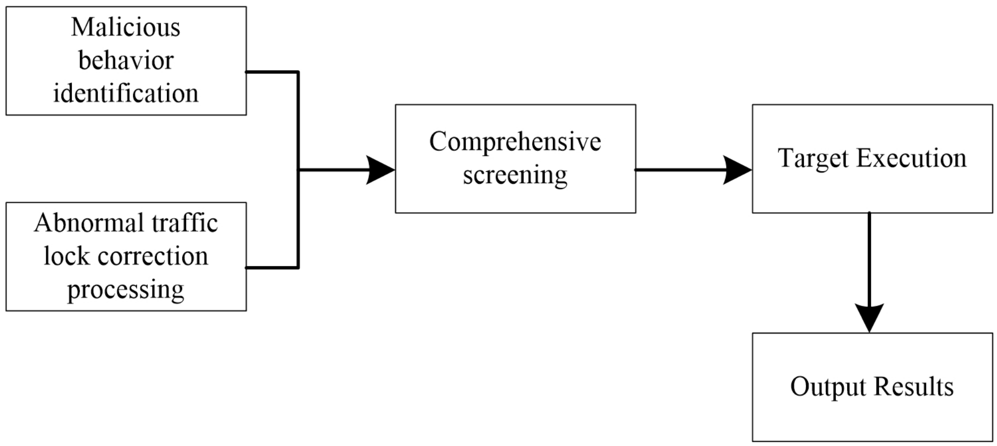

Figure 2: Structure diagram of malicious behavior identification and abnormal traffic locking correction.

Table 1: Basic Test Indicators and Parameter Settings Table.

Name of basic indicators for traffic anomaly detection	Directional measurement reference value	Measured edge measurement reference value
Abnormal detection sequence overlap difference	2.16	1.03
Exception recognition window rules	Identify the target+identify the content+identify the range+calibrate the abnormal position	Identify the target+identify the content+identify the range+calibrate the abnormal position+limit the rules
Hide detection unit values	16.35	18.44
Types of malicious behavior attacks	Eavesdropping attacks, malware attacks, Trojan attacks, vulnerability based attacks	Eavesdropping attacks, malware attacks, Trojan attacks, vulnerability based DDoS attacks
Identifiable range fluctuation ratio	3.51	4.15
Abnormal flow transmission deviation	1.13	1.25

is built for the selected seven cycles, and then, based on this, the fusion depth Q network is used to overlap with the running network to form a stable and circular processing environment. Next, the current traffic situation is controlled and grasped by fusing the deep Q network to form the basic fuzzy processing result. Using self-encoder technology, a multi-level abnormal flow detection framework is constructed, as shown in Figure 3 below:

According to Figure 3, the design and practical analysis of the multi-level abnormal flow detection and testing framework structure are completed. Next, control the abnormal traffic in the first cycle and set up a confusion matrix. At present, it should be noted that malicious behavior can be introduced into the control program first to calculate the predictive value standard of abnormal traffic detection. As shown in Formula 3 below:

$$N = \left(\sum_{W=1} \mathfrak{R}W + \gamma\right)^2 \times \pi W \tag{3}$$

In Formula 3:Nindicates the prediction value standard of abnormal flow detection,$\mathfrak{R}$indicate that reference value of the fusion depth control,Windicates the control frequency,γindicates repeated identification of traffic,πrepresents the controllable stacking range. Combined with the current measurement, the prediction standard of abnormal flow detection is calculated. Next, set the benchmark value for abnormal traffic detection, verify and measure the traffic running state within the period, formulate a combination form of DDoS attack+Trojan attack+vulnerability attack for simulation measurement, detect abnormal network traffic in three stages, and calculate the F1 value, as shown in the following Formula 4:

$$K = \sqrt{(1+h) - \delta\eta} + \sum_{y=1}^{2} hy^2 \tag{4}$$

In Formula 4:Krepresents the F1 value,$h$represents a controllable identification range, mainly measuring attack traffic status through unit attack area to determine controllable identification range,δindicates the unit attack area,ηindicates the attack frequency,$y$indicates the number of attack points. Combined with the current measurement, the analysis of the test results is completed, as shown in the following table 2

According to Table 2, the analysis of the test results is completed: compared with the selected seven cycles of traffic operation, the final F1 value can reach 0.8, which is relatively close to 1. This shows

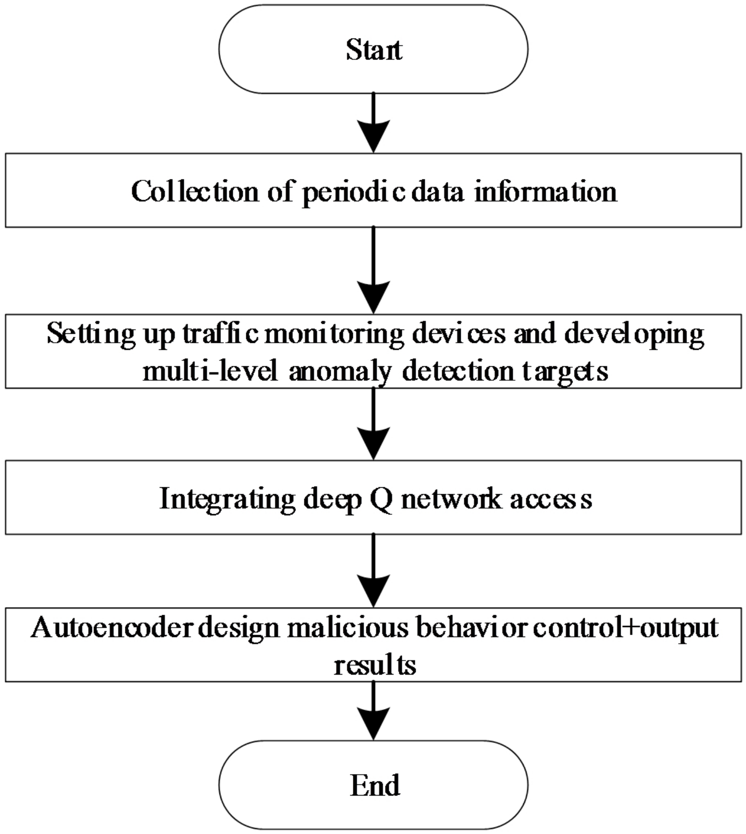

Figure 3: Structural diagram of multi-level abnormal flow detection test framework.

Table 2: Comparative Analysis Table of Test Results.

Abnormal traffic detection testing cycle	Detection time/s	Abnormal conversion difference	F1 value
Test Cycle 1	0.21	1.13	0.851
Test Cycle 2	0.25	1.02	0.863
Test Cycle 3	0.23	1.06	0.891
Test Cycle 4	0.25	1.15	0.857
Test Cycle 5	0.27	1.18	0.871
Test Cycle 6	0.21	1.17	0.833
Test Cycle 7	0.22	1.09	0.864

that with the help of the fusion of deep Q network and self-encoder technology, even in the environment of malicious attack control, the detection of abnormal traffic problems is relatively accurate, the error is controllable, and it has practical application value.

4 CONCLUSION

The overall traffic detection form and malicious behavior simulation structure designed by combining deep Q-network and autoencoder are more flexible, targeted, and stable, which can further solve the problems of low network traffic anomaly detection rate and error control to a certain extent. Expand the range of traffic detection from multiple perspectives, calibrate corresponding abnormal traffic, and facilitate subsequent preprocessing. At the same time, for malicious behavior in the network, a corresponding supply type identification program can be designed in conjunction with the deep Q network, and it can be set in the current control structure to form a larger scale behavior dataset, such as strengthening overall control performance, better solving the imbalance of network traffic, creating a stable and secure network environment, and strengthening the dimension and encryption of personal information, Promoting relevant technologies to a new level of development.

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